

## **Project 3**

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## **Executive Summary**

### **Introduction to the Dataset**

Air pollution is a major environmental and public health issue, affecting millions of people across the United States. Pollutants like Ozone ( $O_3$ ), Carbon Monoxide (CO), Sulfur Dioxide ( $SO_2$ ), and Nitrogen Dioxide ( $NO_2$ ) can contribute to respiratory problems, heart disease, and overall poor air quality. To better understand these pollutants and their impact, this dataset compiles over 600,000 air quality records collected across various locations in the U.S. from 2000 to 2023.

The dataset provides a comprehensive look at pollution trends by including date, location, and detailed pollutant measurements. It tracks air quality over time with year, month, and day values, while geographic details like state, county, city, latitude, and longitude allow for regional comparisons. Each record also contains pollutant concentration levels, including mean values, peak measurements, first maximum occurrence hours, and corresponding Air Quality Index (AQI) scores for each pollutant.

A preliminary analysis shows significant variation in air quality across different locations and time periods. For example,  $O_3$  AQI values range from 0 to 237, with an average of 39.34, while CO AQI values range from 0 to 201, averaging 5.41. Similarly,  $SO_2$  AQI ranges from 0 to 200 (average 5.37) and  $NO_2$  AQI ranges from 0 to 153 (average 22.66). These numbers highlight the diversity of air quality conditions across the country, showing areas with both good and poor air quality.

This dataset is a valuable resource for analyzing pollution patterns, identifying high-risk areas, and understanding long-term trends. By using this data, researchers and policymakers can make informed decisions to improve air quality and reduce pollution-related health risks.

### **Data Delivery**

To extract meaningful insights from the stored data, SQL queries were used to compute pollution trends across different states. One key query focused on analyzing the average O<sub>3</sub> AQI levels across states to determine the regions with the highest pollution levels. The results were then sorted and limited to the top ten states with the highest O<sub>3</sub> AQI values. To facilitate geospatial pollution analysis, the dataset was structured to include latitude and longitude coordinates alongside pollutant concentration values. A dedicated SQL query retrieved relevant pollution metrics, ensuring efficient integration of location-based data for visualization.

### **Data Visualization**

#### **Bar Chart Representation of O<sub>3</sub> AQI Trends**

A horizontal bar chart was generated using Matplotlib to visualize state-level variations in O<sub>3</sub> AQI. The visualization provides a comparative analysis of the ten states with the highest average O<sub>3</sub> AQI, offering a clear representation of pollution intensity across different regions.

#### **Geospatial Heatmap for Pollution Distribution**

A geospatial heatmap was developed using Folium to map the distribution of O<sub>3</sub> AQI across the United States. This visualization technique highlights pollution hotspots by applying a

heatmap layer to latitude and longitude coordinates extracted from the dataset. The heatmap enhances spatial pattern recognition, making it a valuable tool for identifying high-risk areas and informing environmental policy decisions.

### **Sunburst Charts:**

The first Sunburst Chart uses Plotly to display the top 10 U.S. states with the highest average Ozone AQI levels. The chart is color-coded based on AQI values, providing a clear visual representation of air quality across states. The second Sunburst Chart visualizes the average AQI levels for four pollutants (Ozone, Nitrogen Dioxide, Sulfur Dioxide, and Carbon Monoxide) by state. This hierarchical chart breaks down AQI values by pollutants, allowing for an easy comparison of pollution levels across states and pollutants.

### **Heatmap:**

The heatmap is generated using the folium library to display the geographic distribution of Ozone AQI levels across the United States. The map uses latitude and longitude data from the locations table and overlays the AQI data from the pollutants table. The heatmap's intensity is represented through color gradients, offering a visual cue of higher or lower pollution levels across the U.S.

Together, these visualizations provide a comprehensive view of air quality trends, helping to highlight geographic patterns, state-level variations, and fluctuations in pollutant concentrations.

### **Back-end Web Design**

The application is built using the Flask framework, with SQLAlchemy serving as the Object-Relational Mapping (ORM) library to interact with an SQLite database. The design

includes multiple routes, data models, and query functions for filtering and aggregating pollutant data across different geographical locations and timeframes.

### **Flask Framework and Database Configuration**

The web application is implemented using Flask, a lightweight Python web framework. In the `app.py` file, the Flask class is instantiated, initializing the Flask application. The database connection is configured using the `bind_database` function, which connects the application to an SQLite database located at `/home/leonardojrr/usa_pollution_app/usa_pollution.db`. The database configuration includes settings to disable modification tracking and to specify the URI for the SQLite database.

### **Routes and Blueprint Registration**

The web application is modularized using four blueprints: `pollutant_bp`, `states_bp`, `counties_bp`, and `cities_bp`. These blueprints each handle routes related to different aspects of the pollution data, such as pollutants, states, counties, and cities. All blueprints are registered with the main Flask application, with the `url_prefix='/api'` setting ensuring that all routes are prefixed with `/api`, providing a consistent API structure for external consumers.

### **Data Models**

The back-end system uses two primary models to represent location and pollutant data: Location and Pollutant.

- The Location model stores geographical information, including state, county, city, latitude, and longitude. Additionally, the model contains methods to retrieve unique

states, counties, and cities from the database. These methods use SQLAlchemy queries and pandas for data transformation.

- The Pollutant model stores air quality data, such as AQI (Air Quality Index), mean values, maximum values, and first maximum hours for pollutants including Ozone (O3), Carbon Monoxide (CO), Sulfur Dioxide (SO2), and Nitrogen Dioxide (NO2). The pollutants are linked to specific locations via a relationship between the Pollutant and Location models. The Pollutant model includes methods for retrieving pollutant data with various filters, such as date, pollutant type, and AQI.

### **Search and Query Logic**

The application implements advanced querying capabilities, allowing users to filter pollution data by various parameters such as state, county, city, latitude, longitude, year, month, and pollutant values (e.g., AQI, mean, max values). The `get_pollutants` method, for example, allows users to search for pollutant data by these filters and supports querying for specific time periods or pollutant types. The query logic is flexible, allowing users to request pollutant data tailored to their needs.

### **Data Aggregation and Output**

The Pollutant model includes aggregation methods that return pollutant data summaries. For example, methods like `get_top_cleanest_cities` and `get_top_dirtiest_cities` return the top 10 cities with the cleanest and dirtiest air quality, respectively, based on average AQI values. The `get_timeline` method provides data trends for pollutants over time, offering insights into how air quality changes on a broader scale.

The back-end design of this web application leverages Flask and SQLAlchemy to create a scalable and efficient system for querying and serving air quality pollution data. The use of modular blueprints for route handling, combined with powerful ORM models for database interaction, enables flexible and efficient querying. By providing the ability to filter, aggregate, and visualize pollutant data across various locations and timeframes, the application supports the analysis of air quality data in a comprehensive and user-friendly manner.

## **Website**

The front-end of the web application is built using React, with Vite as the build tool for optimized performance. React-DOM is used for efficient rendering and client-side routing. The

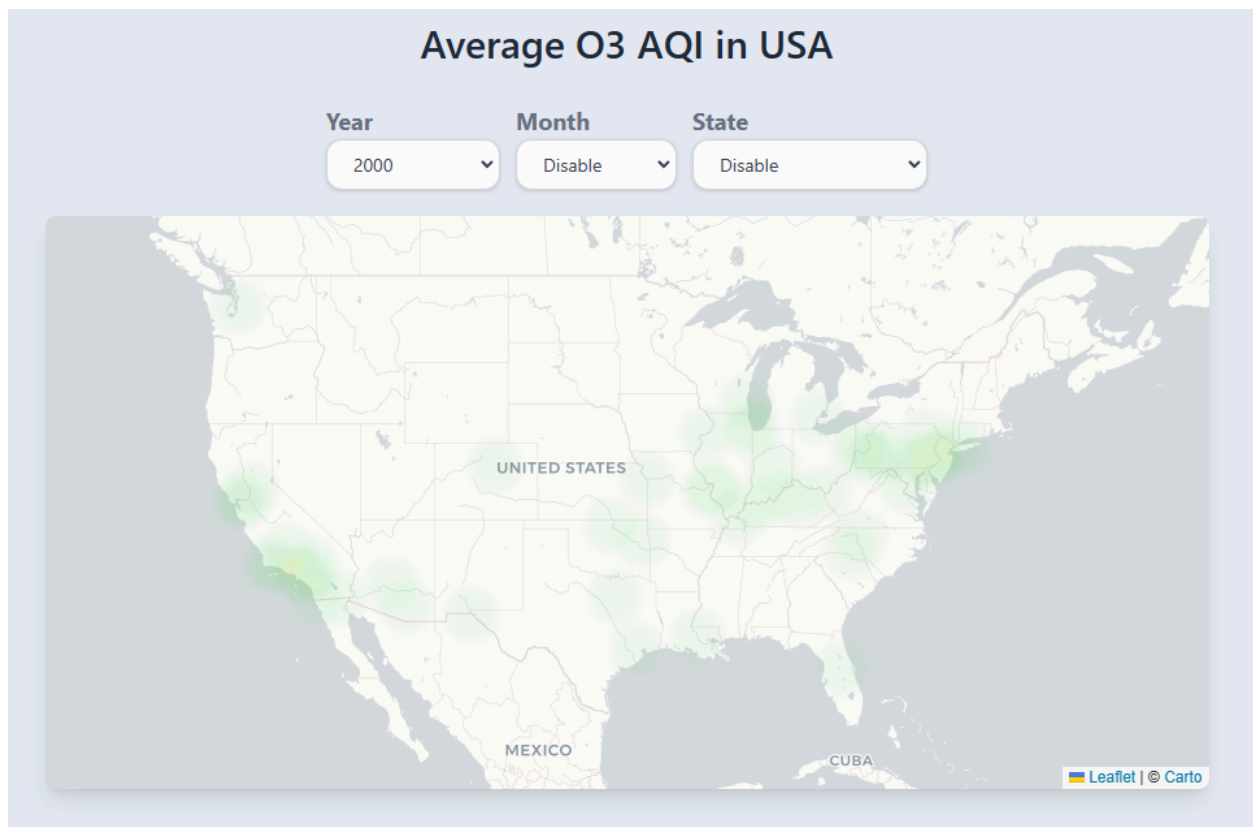
application includes four main routes:

- /home – The landing page providing an overview of the project.
- /pollutants – Displays pollutant data with interactive visualizations.
  - /pollutants/o3 – Ozone-related air quality data.
  - /pollutants/so2 – Sulfur dioxide levels and trends.
  - /pollutants/no2 – Nitrogen dioxide concentrations.
  - /pollutants/co – Carbon monoxide statistics.
- /references – Lists data sources and methodologies.
- /about-us – Information about the project and contributors.

Tailwind CSS is utilized for styling, providing a clean and consistent design with utility-first classes that streamline development.

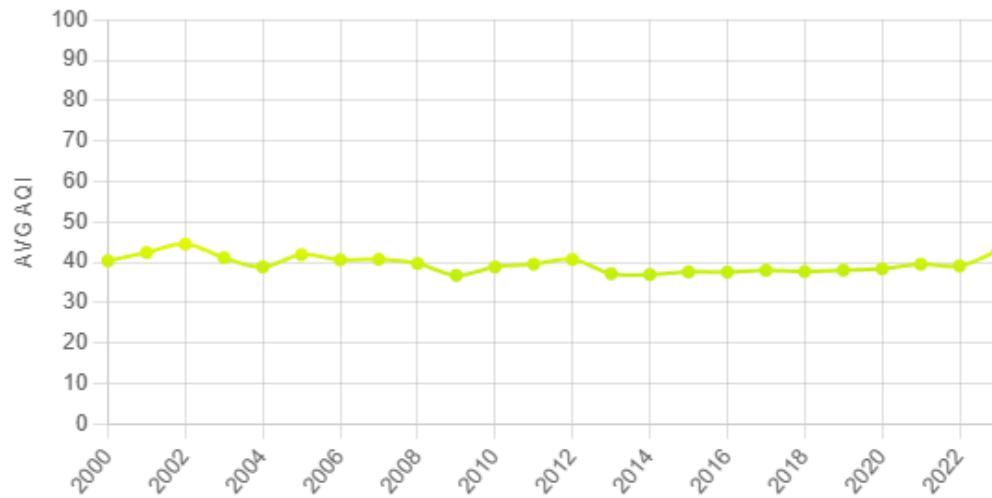
To handle API communication efficiently, an `HttpClient` class was implemented using `Axios`. This class centralizes API calls, improving code maintainability and making it easier to manage requests, error handling, and response processing.

For data visualization, the application leverages `Leaflet` and `Leaflet-heat`, enabling users to interact with a heat map representation of air quality across the USA. Additionally, `Chart.js` and `react-chartjs-2` are used to create interactive bar charts, helping users analyze pollutant levels through dynamic and visually engaging data representations.

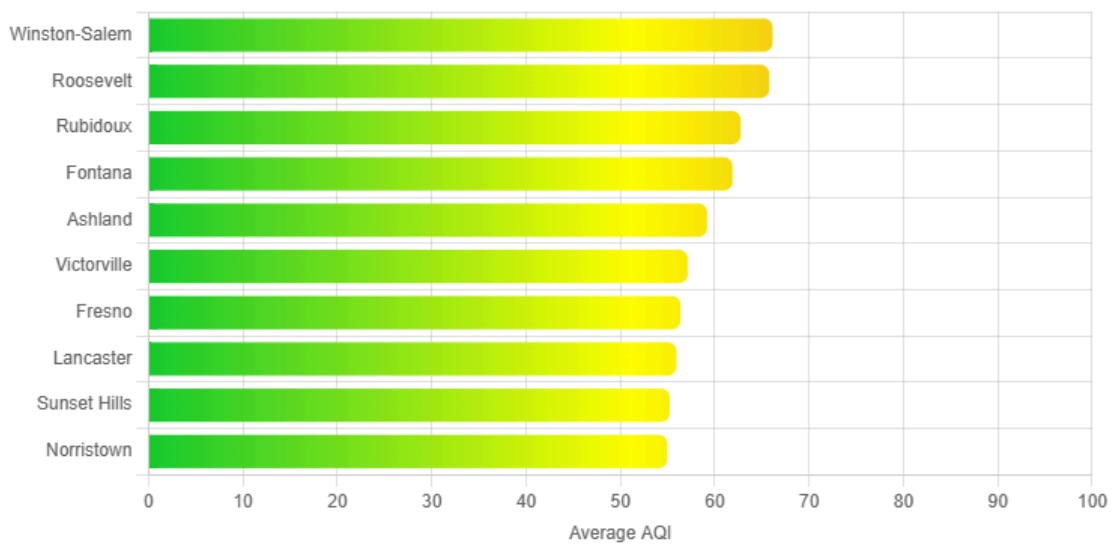




## Average O3 AQI Timeline



## Cities with the Worst O3 AQI



The website provides a user-friendly interface for exploring pollutant data, filtering by location and time, and viewing trends over time. The interactive visualizations help users quickly identify regions with high and low air quality, making complex data more accessible. The efficient API integration with the Flask back end enables real-time data retrieval and updates.

### **Future Work, Bias, and Limitations of Data Set**

In future research, several avenues can be explored to build upon the findings of this study. Firstly, expanding the dataset to include more recent data and additional pollutants could provide a more comprehensive understanding of air quality trends. Integrating data from other sources, such as satellite imagery and ground-based sensors, could enhance the accuracy and granularity of the analysis. Additionally, developing predictive models using machine learning techniques could help forecast pollution levels and identify potential high-risk areas. Collaborating with policymakers and environmental agencies to implement and evaluate intervention strategies based on the findings could also be a valuable direction for future work.

### **Bias**

The dataset used in this study may contain inherent biases that could affect the results. For instance, the data collection methods and locations may not be uniformly distributed across all regions, leading to potential overrepresentation or underrepresentation of certain areas.

Additionally, the accuracy of the pollutant measurements may vary depending on the equipment and protocols used at different monitoring stations. It is important to acknowledge these biases and consider their potential impact on the findings when interpreting the results.

### **Limitations of the Data Set**

There are several limitations associated with the dataset used in this study. Firstly, the dataset covers a specific time period (2000 to 2023) and may not capture long-term trends or recent changes in air quality. Secondly, the dataset primarily focuses on four pollutants (Ozone, Carbon Monoxide, Sulfur Dioxide, and Nitrogen Dioxide), which may not provide a complete picture of overall air quality. Thirdly, the spatial resolution of the data may be limited, as it relies on monitoring stations that may not be evenly distributed across all regions. Finally, the dataset does not account for other factors that could influence air quality, such as meteorological conditions, population density, and industrial activities. These limitations should be considered when interpreting the results and drawing conclusions from the analysis.

## Conclusion

This study provides a comprehensive analysis of air pollution trends in the United States, focusing on key pollutants such as Ozone (O<sub>3</sub>), Carbon Monoxide (CO), Sulfur Dioxide (SO<sub>2</sub>), and Nitrogen Dioxide (NO<sub>2</sub>). By leveraging a robust data set spanning from 2000 to 2023, we have identified significant variations in air quality across different regions and time periods. The use of SQL queries and advanced data visualization techniques has enabled us to extract meaningful insights and highlight pollution hotspots.

Our findings underscore the importance of continuous monitoring and analysis of air quality data to inform policy decisions and mitigate pollution-related health risks. The integration of geospatial analysis and web-based data delivery systems further enhances the accessibility and usability of the data for researchers, policymakers, and the public.

While this study has provided valuable insights, it is important to acknowledge the limitations and biases inherent in the dataset. Future research should aim to address these limitations by incorporating more recent data, additional pollutants, and integrating data from diverse sources. By doing so, we can develop more accurate predictive models and implement effective intervention strategies to improve air quality and protect public health.

Overall, this study contributes to the growing body of knowledge on air pollution and provides a solid foundation for future research and policy development in this critical area.